

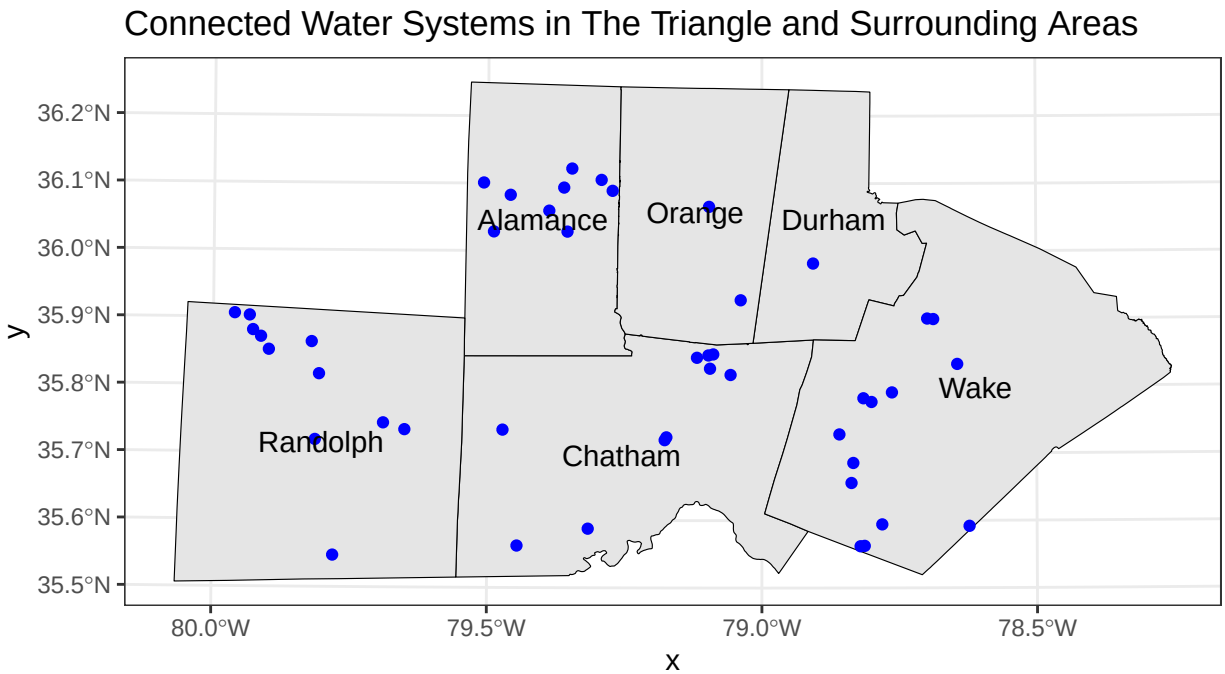
Plan 372 HW5

Lily Monte

2025-04-08

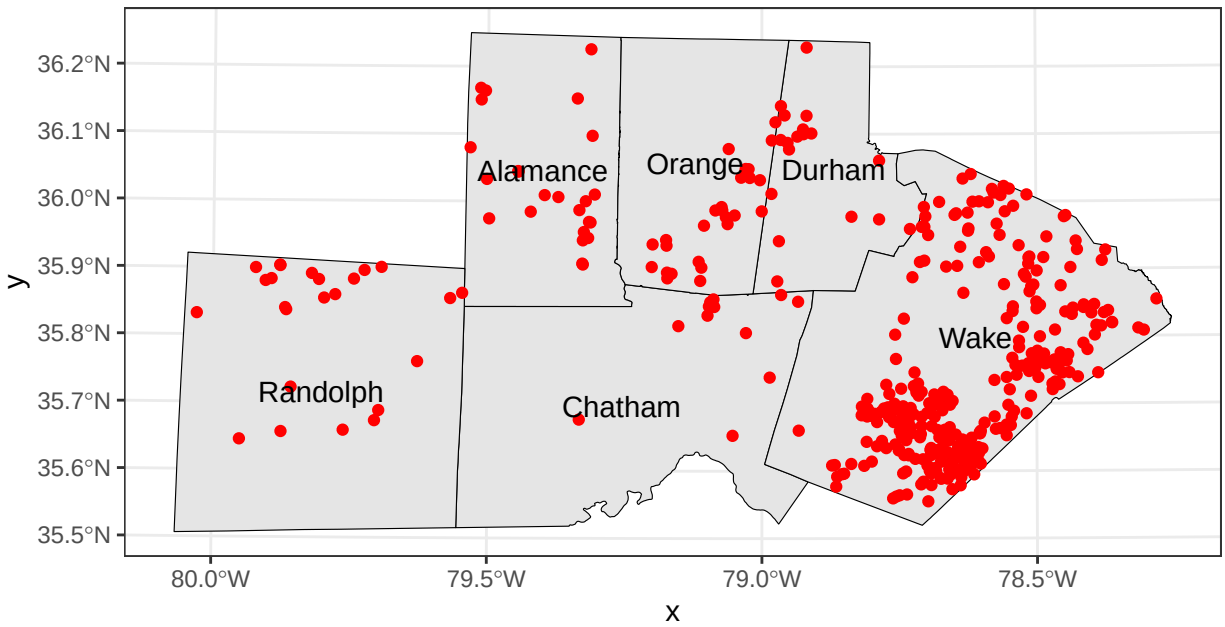
Introduction

The Triangle and surrounding Raleigh-Durham-Cary area includes several water systems, some of which purchase water from each other to meet the demands of their larger populations. As depicted in the following map, There are several connected water systems, mostly concentrated in Wake, Alamance, and Randolph county.



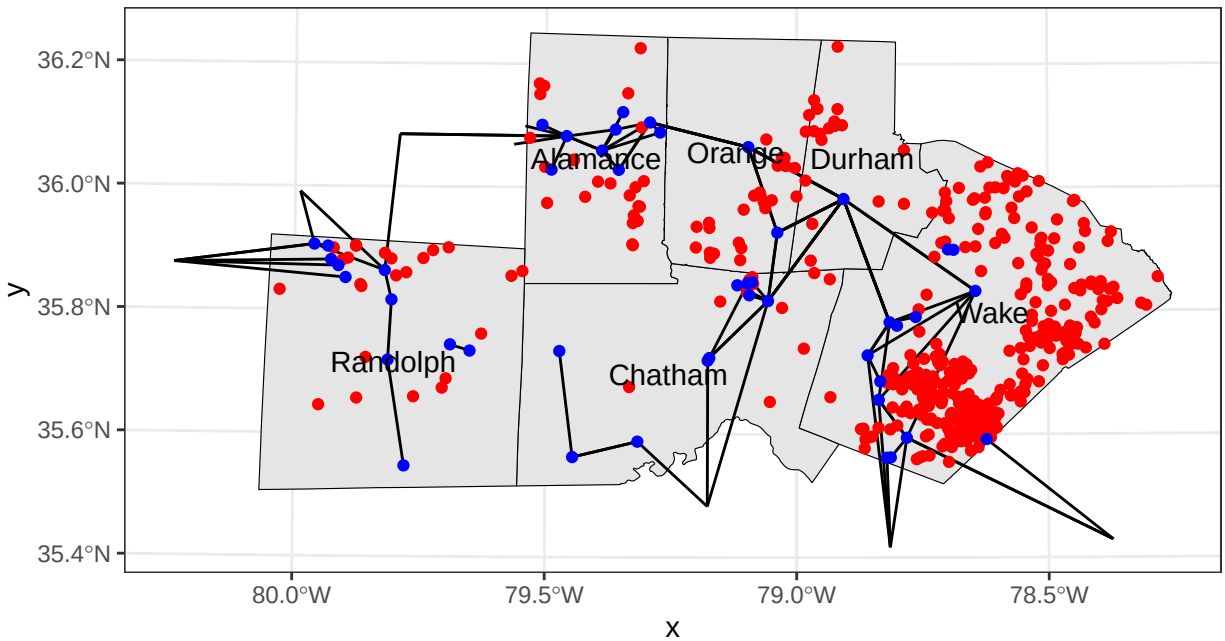
Conversely, the vast majority of water systems in the region are unconnected, and many of them are small-scale, often servicing only a neighborhood or trailer park. Some of them service entire small towns or cities, as well.

Unconnected Water Systems in The Triangle and Surrounding Areas



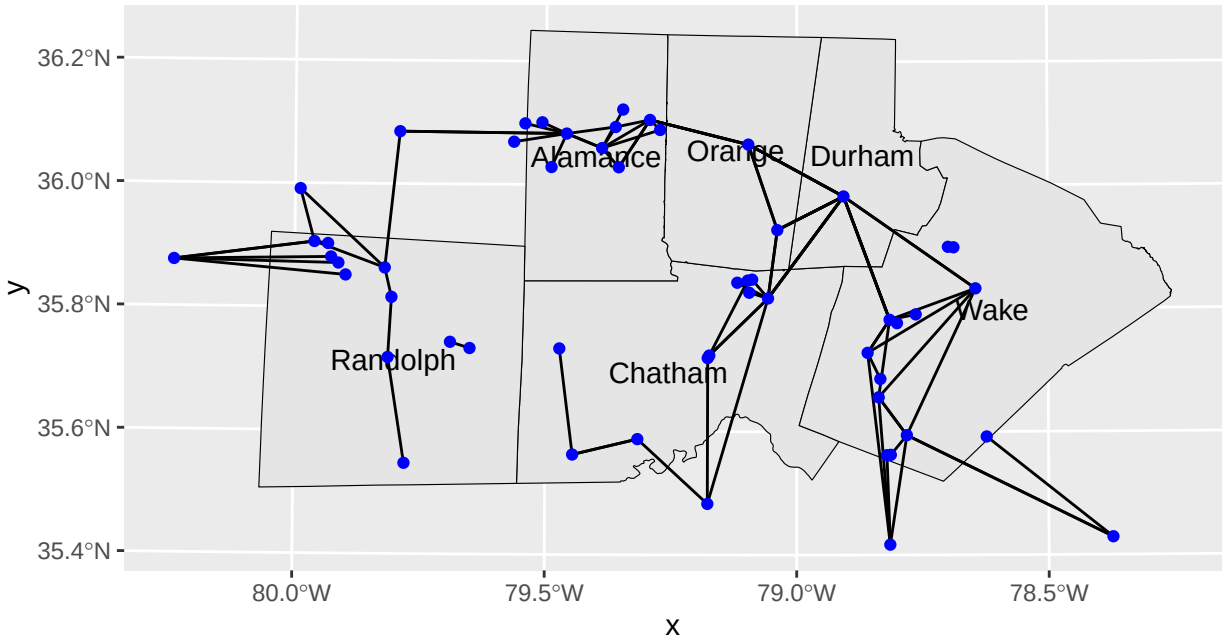
Lastly, the following graph depicts all connected water systems (in blue), all unconnected systems (in red), and the connections between them. There are a few water systems connecting to systems outside of the dataset, which are not represented by a point in this graph.

All Water Systems in The Triangle and Surrounding Areas



Graph theory allows researchers to depict complex networks of interactions and infrastructure as vertices and edges connecting them. The below graph is a network representation of connected water infrastructures in the Triangle and Raleigh-Durham-Cary areas.

```
## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'nodeID = group_indices(., factor(xy, levels = unique(xy)))'.
## Caused by warning:
## ! The '...' argument of 'group_indices()' is deprecated as of dplyr 1.0.0.
## i Please 'group_by()' first
```



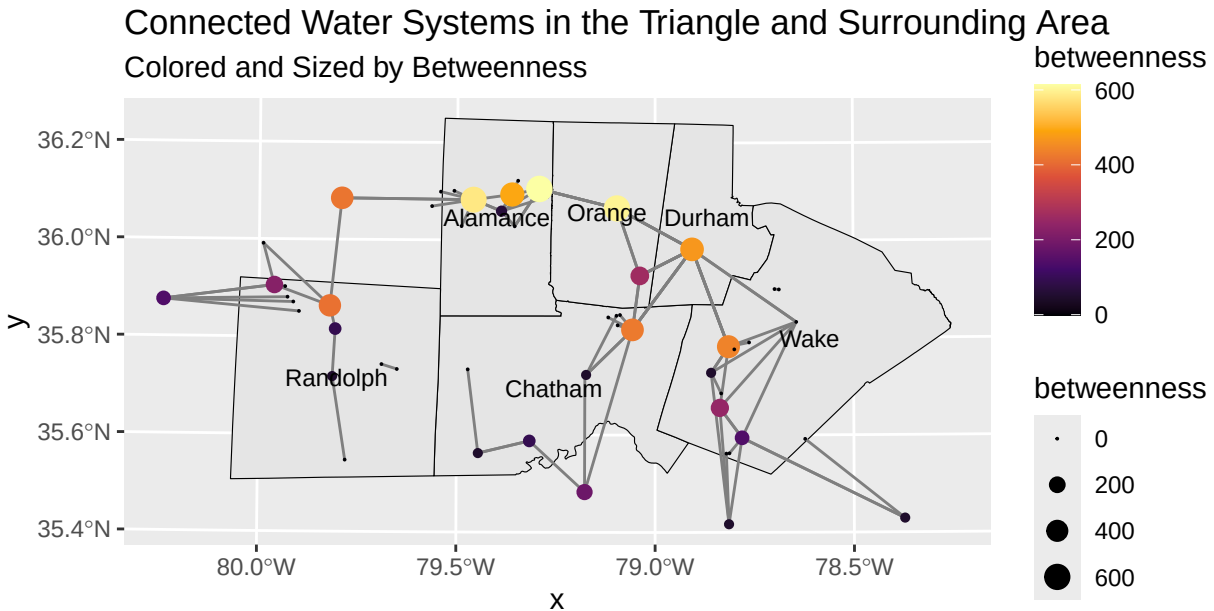
```

## # A tbl_graph: 54 nodes and 91 edges
## #
## # An undirected multigraph with 3 components
## #
## # Node Data: 54 x 2 (active)
##   nodeID      geometry
##   <int>      <POINT [°]>
## 1         1 (-79.3885 36.0594)
## 2         2 (-79.4591 36.0827)
## 3         3 (-79.2728 36.0891)
## 4         4 (-79.3614 36.0937)
## 5         5 (-79.3467 36.122)
## 6         6 (-79.293 36.1053)
## 7         7 (-79.3551 36.0284)
## 8         8 (-79.7917 36.0849)
## 9         9 (-79.4894 36.0283)
## 10        10 (-79.542 36.0989)
## # i 44 more rows
## #
## # Edge Data: 91 x 7
##   from    to Name      Description capacity      geom edgeID
##   <int> <int> <chr>      <chr>      <chr>      <LINESTRING [°]> <int>
## 1      1     2 Burlington,~ BuyerID: N~ Contrac~ (-79.3885 36.0594, -79.4~      1
## 2      1     3 Mebane, Tow~ BuyerID: N~ <NA>      (-79.3885 36.0594, -79.2~      2
## 3      1     4 Haw River, ~ BuyerID: N~ Contrac~ (-79.3885 36.0594, -79.3~      3
## # i 88 more rows

```

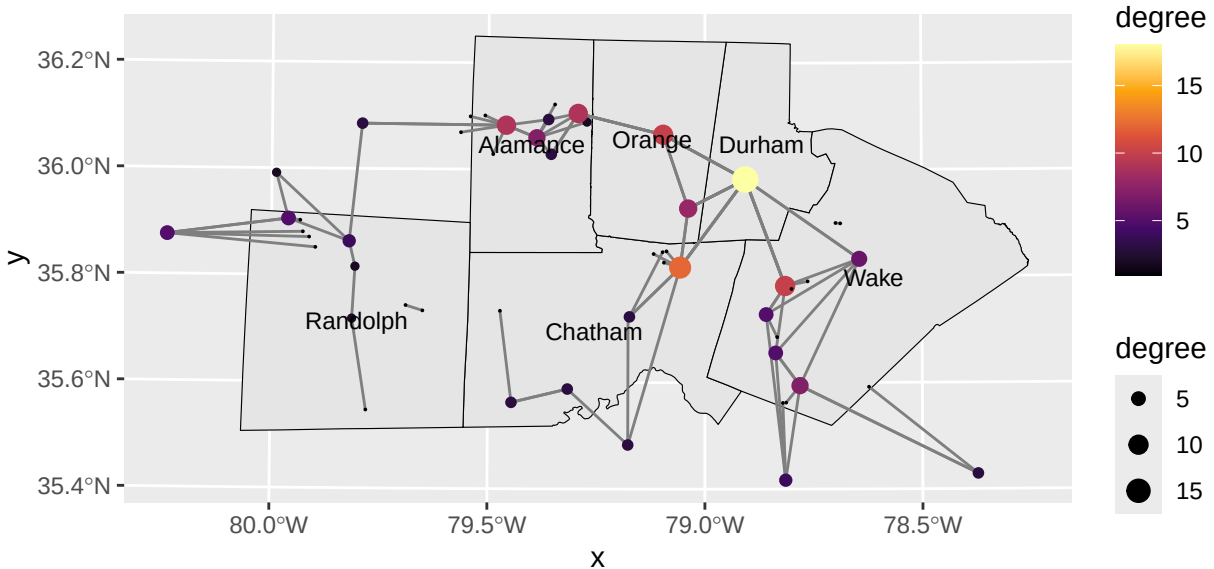
#Centrality Measures

To understand how these water systems interact, centrality measures were conducted. First, we calculated betweenness, indicating the number of paths which potentially pass through the system.



Contrary to what one might expect, the cities with the highest betweenness were Hillsborough and Mebane, two relatively small cities, especially considering the presence of much larger cities such as Durham and Raleigh which might have greater capacity due to infrastructure or greater need due to high populations. As expected, areas situated nearer urban cores had higher betweenness compared to rural areas; it is likely that these rural areas simply purchase water from one system, whereas larger cities buy and sell water capacity to various other outlying areas.

Connected Water Systems in the Triangle and Surrounding Area Colored and Sized by Degree Centrality



The above map demonstrates degree centrality, which simply measures how many connections each water system has. As expected, Durham, one of the largest cities in the triangle and the state at large, has the highest degree centrality, with over 15 different connections. The implications of this are fairly apparent, and indicate that much of the water infrastructure in the triangle area is dependent on the infrastructure of one city. This puts the water infrastructure of the region in a precarious position; if the water infrastructure in Durham were to fail due to environmental catastrophe, for example, much of the rest of the triangle would be vulnerable to water issues. This is not to say that trading water is a poor choice, but rather that concentrating all infrastructure in a few locations could prove a risk.

```
## Simple feature collection with 46 features and 2 fields
## Geometry type: POINT
## Dimension: XY
## Bounding box: xmin: -79.9611 ymin: 35.5472 xmax: -78.62274 ymax: 36.122
## Geodetic CRS: WGS 84
## First 10 features:
##      Name
## 1  Burlington, City Of
## 2    Graham, City Of
## 3    Mebane, Town Of
## 4  Haw River, Town Of
## 5      Elon, Town Of
## 6 Green Level, Town Of
## 7 Village Of Alamance
## 8  Asheboro, City Of
## 9  Randleman, City Of
## 10 Ramseur, Town Of
```

```

##
## 1 PWSID: NC0201010<br>Source_Water_Type: SW<br>Owner_Type: Local Government<br>Service_Population_E
## 2 PWSID: NC0201015<br>Source_Water_Type: SW<br>Owner_Type: Local Government<br>Service_Popula
## 3 PWSID: NC0201018<br>Source_Water_Type: SWP<br>Owner_Type: Local Government<br>Service_Population_E
## 4 PWSID: NC0201020<br>Source_Water_Type: SWP<br>Owner_Type: Local Government<br>Service_Population_E
## 5
## 6
## 7
## 8 PWSID: NC0276010<br>Source_Water_Type: SWP<br>Owner_Type: Local Government<br>Service_Population_E
## 9
## 10
## geometry
## 1 POINT (-79.4591 36.0827)
## 2 POINT (-79.3885 36.0594)
## 3 POINT (-79.2728 36.0891)
## 4 POINT (-79.3614 36.0937)
## 5 POINT (-79.508 36.1008)
## 6 POINT (-79.3467 36.122)
## 7 POINT (-79.4894 36.0283)
## 8 POINT (-79.8136 35.7186)
## 9 POINT (-79.8069 35.8162)
## 10 POINT (-79.6508 35.7341)

## [[1]]
## + 3/54 vertices, from 7e9326a:
## [1] 31 30 32

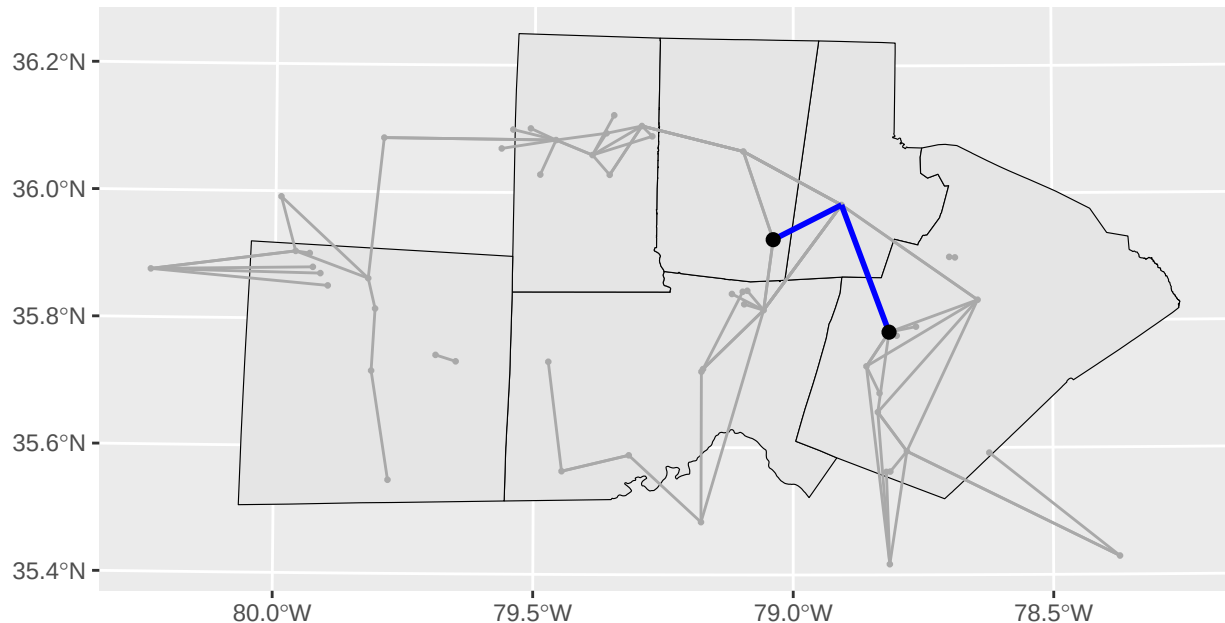
## Warning: 'subgraph.edges()' was deprecated in igraph 2.1.0.
## i Please use 'subgraph_from_edges()' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

## # A tbl_graph: 3 nodes and 2 edges
## #
## # An unrooted tree
## #
## # Node Data: 3 x 4 (active)
## nodeID geometry degree betweenness
## <int> <POINT [°]> <dbl> <dbl>
## 1 30 (-78.9067 35.9815) 18 468
## 2 31 (-79.0387 35.927) 8 264
## 3 32 (-78.8152 35.7815) 10 438
## #
## # Edge Data: 2 x 9
## from to Name Description capacity geom edgeID length
## <int> <int> <chr> <chr> <chr> <LINESTRING [°]> <int> [m]
## 1 1 2 Durh~ BuyerID: N~ Contrac~ (-79.0387 35.927, -78.90~ 41 13338.
## 2 1 3 DURH~ BuyerID: N~ Contrac~ (-78.8152 35.7815, -78.9~ 55 23718.
## # i 1 more variable: betweenness <dbl>

## # A tibble: 1 x 1
## length

```

```
##      [m]
## 1 37055.
```



Proposals for OWASA’s Purchase of Water

The above map details the route between the City of Cary water system and the OWASA system. To purchase water from Cary, OWASA would need to route through existing infrastructure in Durham, which is a distance of 37055 meters, or about 23 miles. This would be significantly cheaper than purchasing from Raleigh, which is roughly 26 miles away when routed through Durham, as seen in the below tables and graphs. In both cases, the water would only be routed through Durham, and the difference in distance is small, but depending on the cost per mile, could be very significant in deciding from whom to purchase water.

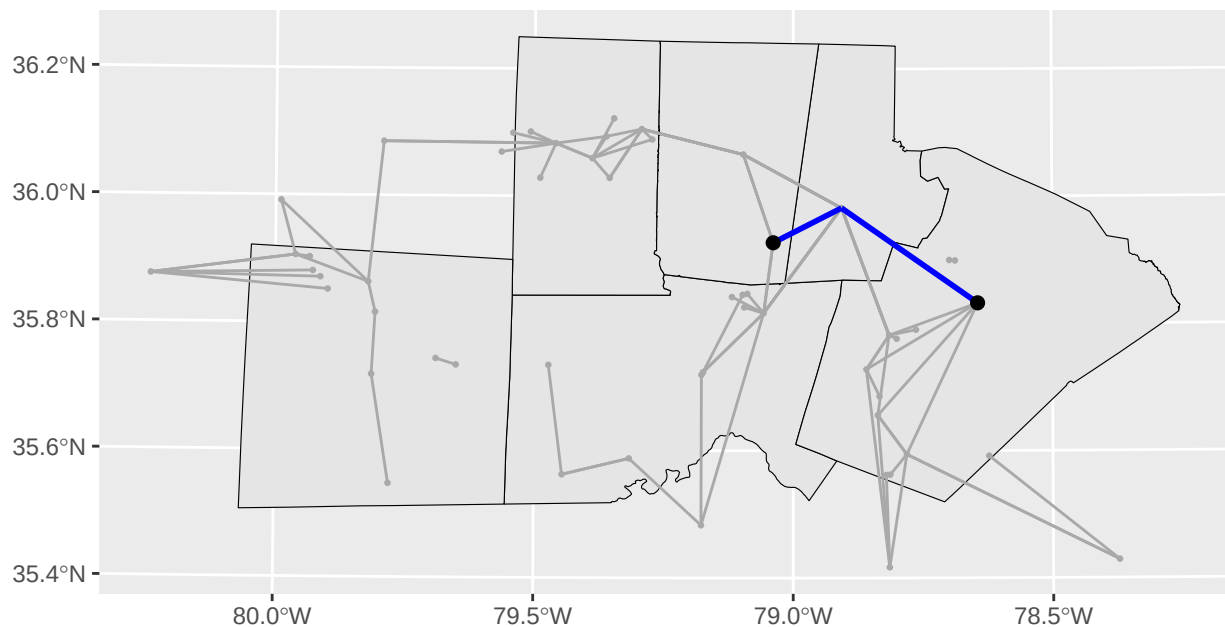
```
## [[1]]
## + 3/54 vertices, from 7e9326a:
## [1] 31 30 37

## # A tbl_graph: 3 nodes and 2 edges
## #
## # An unrooted tree
## #
## # Node Data: 3 x 4 (active)
##   nodeID      geometry degree betweenness
##   <int>      <POINT [°]> <dbl>      <dbl>
```



```
## 1      30 (-78.9067 35.9815)      18      468
## 2      31 (-79.0387 35.927)       8      264
## 3      37 (-78.6442 35.8321)       6       0
## #
## # Edge Data: 2 x 9
##   from   to Name Description capacity      geom edgeID length
##   <int> <int> <chr> <chr>      <chr>      <LINESTRING [°]> <int>    [m]
## 1      1     2 Durh~ BuyerID: N~ Contrac~ (-79.0387 35.927, -78.90~      41 13338.
## 2      1     3 DURH~ BuyerID: N~ <NA>      (-78.6442 35.8321, -78.9~      51 28895.
## # i 1 more variable: betweenness <dbl>

## # A tibble: 1 x 1
##   length
##   [m]
## 1 42233.
```



Liberty, NC

The town of Liberty is not connected to an existing water network. However, were the municipal government of Liberty to purchase water, they would likely wish to do so from the closest node in the broader connected network. The closest node in the network is the town of Ranseur, but this presents an issue; as a small town with less than 2,000 residents, it likely does not have the infrastructure available to supply water to the town of Liberty. Therefore, I calculated the closest node with a degree centrality of 5, which is the city of Burlington. This is likely a better choice as the city will more than likely have a more capable infrastructure

and an ability to sell excess water. With this in mind, however, it still may be more viable to connect to the Ransour network, as it would require fewer costs in terms of creating new infrastructure for water trading.

```
#Theres probably a better way to do this but this takes the coordinates of liberty, nc, and assigns the
liberty = st_point(c(-79.5682, 35.8557)) %>%
  st_sfc(crs = 4326)

#This puts those coordinates into a matrix
coords_origin <- liberty %>%
  st_coordinates() %>%
  matrix(ncol = 2)

#The following two blocks of codes convert the nodes (which are in dataframe format) into spatial data
nodes <- graph %>%
  activate(nodes) %>%
  as_tibble() %>%
  st_as_sf()

coords <- nodes %>%
  st_coordinates()

#The following three lines of code finds the closest coordinates in Coords to coords_origin, then index
node_index_o <- knn(data = coords, query = coords_origin, k = 1)
node_o <- nodes[node_index_o$nn.idx, ]

node_o$nodeID
```

```
## [1] 45
```

```
connectednodes$Name[connectednodes$nodeID == node_o$nodeID]
```

```
## [1] "Ramseur, Town Of"
```

```
#This filters for only nodes with high betweenness, then does the same thing.
coords_filtered <- nodes %>%
  filter(degree >= 5) %>%
  st_coordinates()

node_index_o_filtered <- knn(data = coords_filtered, query = coords_origin, k = 1)
node_o_filtered <- nodes[node_index_o_filtered$nn.idx, ]

node_o_filtered$nodeID
```

```
## [1] 2
```

```
connectednodes$Name[connectednodes$nodeID == node_o_filtered$nodeID]
```

```
## [1] "Burlington, City Of"
```